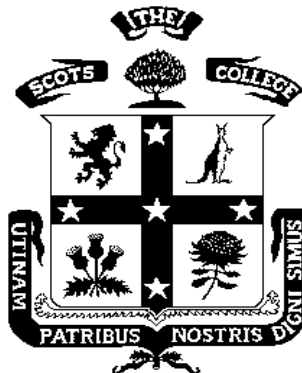


THE SCOTS COLLEGE Sydney



Extension One Mathematics

HSC Task 3

Weighting 20%

1st June 2009

Total marks – 33

- Attempt all questions.

General Instructions

- **Working time – 45 Minutes.**
- Write using black or blue pen.
- Start a new page for every question
- Board-approved calculators may be used.
- All necessary working should be shown in every question.

Table of Standard Integrals provided at the end of the paper.

QUESTION 1**(10 Marks)**

- a) The temperature of a cup of coffee varies according to the rate given by $\frac{dT}{dt} = -k(T - T_o)$, where T is the temperature in degrees after elapsed time, t (in minutes), and T_o is the temperature of the environment.

The cup, initially at 120°C , is kept in a cold chamber at -20°C . After 3 minutes, the temperature of the cup drops down to 80°C .

- i. Show that $T = T_o + Ae^{-kt}$ is a possible function that represents the variation of temperature with time for the cup. [1]
 - ii. Find the values of A and k . [3]
 - iii. If the cup at 80°C is now placed in a room whose temperature is 20°C , assuming that the value of k remains unchanged, find the temperature of the cup after a further 20 minutes. [2]
- b) A kite flying at a *constant* height of 40 m above the ground, is being dragged along by wind at a rate of 10 m/s. The kite is initially vertically above the ground. At what rate is the length of the string, tied to the kite, being released from the ground, increasing after 3 seconds. (Assume that the string remains straight). [4]

QUESTION 2**(14 Marks)****START A NEW PAGE**

- a) Prove the following by mathematical induction

$$2\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{9}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{n} \quad \text{for all positive integers } n \geq 2. \quad [4]$$

- b) Consider the series $\sum_{r=1}^{\infty} (\log_e x)^r$, where $x > 0$. [6]
- Write down the first term, common ratio and the sum of n terms of the series.
 - Find the range of values of x , such that a limiting sum exists for this series.
 - Find the limiting sum if $x = \sqrt{e}$.
- c) Let T_n and S_n represent the n^{th} Term and the Sum of n terms respectively, of an Arithmetic Progression, with first term a and common difference d ($a, d \neq 0$). If T_{10} , T_4 and T_6 form consecutive terms of a Geometric Progression,
- Show that $S_{10} = 0$. [2]
 - Show that $S_6 + S_{12} = 0$ [2]

QUESTION 3

(9 Marks)

START A NEW PAGE

Gordon and Gabbie take a loan of \$500,000 from Community Bank, to buy a new house. The period of the loan is 30 years and interest is charged at the rate of 6% p.a. on the amount owing. Repayment is through a fixed monthly instalment of \$ M , paid at the end of each month.

Let A_n be the amount owing at the end of the n^{th} month, after the payment of the monthly instalment.

- Show that at the end of the third month, the amount owing is given by

$$A_3 = 500000(1.005)^3 - M(1 + 1.005 + 1.005^2) \quad [2]$$

- By first arriving at a general expression for A_n , find the value of M . [2]

- Find the amount owing to the bank at the end of 4 years. [2]

- At the end of the 4th year, the bank raises the interest rate to 7.2%. At the same time, Gordon and Gabbie decide to make fixed monthly payments of \$4200 to the bank. Find the time it would now take for the couple to completely pay off the loan. Express your answer in years and months. [3]

END OF PAPER

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

Yr 12 Maths Ext 1 Task 3
SOLUTIONS

(a) (i) $T = T_0 + Ae^{-kt}$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(T - T_0) \quad (\because Ae^{-kt} = T - T_0)$$

Hence $T = T_0 + Ae^{-kt}$ is a possible solution

(ii) When $t = 0$, $T = 120^\circ\text{C}$, $T_0 = -20^\circ\text{C}$

$$120 = -20 + Ae^0$$

$$\therefore A = 140^\circ\text{C}$$

When $t = 3$ minutes, $T = 80^\circ\text{C}$

$$80 = -20 + 140e^{-3k}$$

$$100 = 140e^{-3k}$$

$$e^{-3k} = \frac{10}{14}$$

$$= \frac{5}{7}$$

$$\therefore -3k = \log_e \frac{5}{7}$$

$$k = -\frac{1}{3} \log_e \frac{5}{7} = 0.112157 \dots$$

$$\approx 0.112$$

(iii) $T_0 = 20^\circ\text{C}$

at $t = 0$, $T = 80^\circ\text{C}$

$$80 = 20 + Ae^0 \quad \therefore A = 60$$

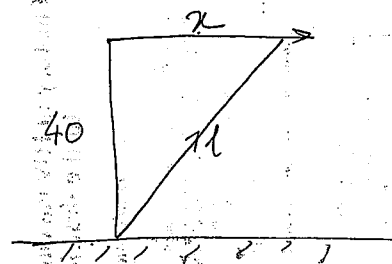
When $t = 15$ min, $T = 20 + 60e^{-15 \times k}$

$$= 31.156 \dots$$

$$= 31.2^\circ\text{C}$$

Question 1

(b)



$$\frac{dx}{dt} = 10 \text{ m/s}$$

$$\frac{dl}{dt} = ?$$

$$l^2 = x^2 + 40^2$$

$$l = \sqrt{x^2 + 1600} \quad (\because l > 0)$$

$$\frac{dl}{dx} = \frac{1}{2} (x^2 + 1600)^{-\frac{1}{2}} \cdot 2x$$

$$= \frac{x}{\sqrt{x^2 + 1600}}$$

$$\frac{dl}{dt} = \frac{dx}{dt} \times \frac{dl}{dx}$$

$$= 10 \times \frac{x}{\sqrt{x^2 + 1600}}$$

When $t = 3$ seconds $x = 30$

$$\frac{dl}{dt} = \frac{10 \times 30}{\sqrt{900 + 1600}}$$

$$= \frac{10 \times 30}{50}$$

$$= 6 \text{ m/s}$$

Question 2

(a) Step 1: prove true for $n=2$

$$\text{LHS: } 2(1 - \frac{1}{4}) = 2 \times \frac{3}{4} = \frac{3}{2}$$

$$\text{RHS: } \frac{2}{2+1} = \frac{2}{3}$$

Hence true for $n=1$

Step 2: Assume true for $n=k$

$$\text{i.e. } 2(1 - \frac{1}{4})(1 - \frac{1}{9}) \dots (1 - \frac{1}{k^2}) = \frac{k}{k+1}, k \geq 2$$

Step 3: Prove true for $k=k+1$

$$\text{i.e. } 2(1 - \frac{1}{4})(1 - \frac{1}{9}) \dots (1 - \frac{1}{(k+1)^2}) = \frac{k+2}{k+1}$$

$$\text{LHS: } \left(\frac{k}{k+1} \right) \left(1 - \frac{1}{(k+1)^2} \right) \quad (\text{from step 2})$$

$$= \frac{k}{k+1} - \frac{k}{(k+1)^2}$$

$$= \frac{1}{(k+1)^2} [k(k+1) - k]$$

$$= \frac{k(k+1)}{(k+1)^2} (k^2 + 2k + 1 - k)$$

$$= \frac{k(k+1)}{k(k+2)}$$

$$= \frac{k+2}{k+1}$$

$$= \text{RHS}$$

Hence true for $n=k+1$

Therefore by the Principle of Mathematical Induction, the statement is true for all $n \geq 2$

(b) Series (i) $a = \log_e x$, $r = \log_e x$, $S_n = \frac{\log_e x (\log_e x)^{n-1}}{\log_e x - 1}$

(ii) $S_\infty = \log_e x + (\log_e x)^2 + (\log_e x)^3 + \dots$

$$|\log_e x| < 1$$

$$\log_e x < 1 \quad \text{or} \quad \log_e x > -1$$

$$x < e^1$$

$$\frac{1}{e} < x < e, \quad \left(\because \log_e 1 = 0 \right)$$

$$(ii) S_\infty = \frac{a}{1-r}$$

$$a = \log_e \sqrt{e} = \frac{1}{2}$$

$$r = \log_e \sqrt{e} = \frac{1}{2}$$

$$S_\infty = \frac{1 - \frac{1}{2}}{\frac{1}{2}} = 1$$

(c) (i)

$$T_6 = a + 5d$$

$$T_4 = a + 3d$$

$$T_{10} = a + 9d$$

$a = \text{first term}$, $d = \text{common difference}$

$$\frac{a+3d}{a+5d} = \frac{a+3d}{a+3d}$$

$$\text{or } (a+3d)^2 = (a+9d)(a+5d)$$

$$a^2 + 6ad + 9d^2 = a^2 + 14ad + 45d^2$$

$$8ad + 36d^2 = 0 \quad \text{or } 4d(2a+9d) = 0$$

$$d \neq 0 \therefore 2a+9d = 0$$

$$\begin{aligned}
 S_{10} &= \frac{10}{2} (2a + 9d) \\
 &= 5 (2a + 9d) \\
 &= 0 \quad (\because 2a + 9d = 0)
 \end{aligned}$$

$$(ii) \quad S_6 + S_{12} = 0$$

$$\begin{aligned}
 LHS &= \frac{6}{2} (2a + 5d) + \frac{12}{2} (2a + 11d) \\
 &= 6a + 15d + 12a + 66d \\
 &= 18a + 81d \\
 &= 9 (2a + 9d) \\
 &= 0 \quad (\because 2a + 9d = 0)
 \end{aligned}$$

Yr 12 Extension 1 Mathematics

Assessment Task 3 - SOLUTIONS

Question 3

(a) Amt Borrowed = \$500,000

Interest Rate: 6% p.a. = 0.5% p.month = 0.005

Period: 30 years = 360 months

Repayment: \$M per month.

(i) A_n = Amt owing at the end of n^{th} month.

$$A_1 = 500000 (1.005) - M$$

$$\begin{aligned}
 A_2 &= [500000 (1.005) - M] 1.005 - M \\
 &= 500000 (1.005)^2 - 1.005M - M
 \end{aligned}$$

$$\begin{aligned}
 A_3 &= 500000 (1.005)^3 - 1.005^2 M - 1.005M - M \\
 &= 500000 (1.005)^3 - M (1 + 1.005 + 1.005^2)
 \end{aligned}$$

$$(ii) \quad A_n = 500000 (1.005)^n - M (1 + 1.005 + 1.005^2 + \dots + 1.005^{n-1})$$

$$0 = A_{360} = 500000 (1.005)^{360} - M (1 + 1.005 + 1.005^2 + \dots + 1.005^{359})$$

$$M \left[\frac{1 (1.005^{360} - 1)}{1.005 - 1} \right] = 500000 (1.005)^{360}$$

$$\therefore M = \frac{0.005 \times 500000 (1.005)^{360}}{1.005^{360} - 1}$$

$$= 2997.7526 \dots$$

$$= \$2997.75$$

(iii) $A_{48} = 500000(1.005)^{48} - 2997.75 \left(\frac{1.005^{48} - 1}{1.005 - 1} \right)$

$= 473072.807 \dots$
 $\approx \$ 473073$

(iv) $r = 7.2\% p.a. = 0.006$
 $M = \$4200, P = \473073

$A_n = 473073(1.006)^n - 4200(1 + 1.006 + 1.006^2 + \dots + 1.006^{n-1})$
 $0 = 473073(1.006)^n - 4200 \left(\frac{1.006^n - 1}{1.006 - 1} \right)$

$= 473073(1.006)^n - 700000(1.006^n - 1)$

$= 473073(1.006)^n - 700000(1.006^n) + 700000$

$226987(1.006)^n = 700000$

$1.006^n = \frac{700000}{226987}$

$n \log 1.006 = \log \frac{700000}{226987}$

$n = \frac{\log \frac{700000}{226987}}{\log 1.006}$

$= 188.3 \text{ months}$

$= 15.7 \text{ years i.e. 15 years } 8^{\text{th}} \text{ month}$

Therefore, they repay the loan in a further 15 yrs 8th month or total of 19 years 8th month.